Hypothalamus · hierarchical baseline DETECTION

New-section p:95% predictive
Anatomy μ:95% credible
Posterior μ mean
Measured pooled fraction

A Tll1

HY-unassigned (n=6,393; S=18)
LHA (n=4,502; S=11)
ZI-unassigned (n=3,518; S=9)
AHN (n=2,308; S=8)
PH (n=1,419; S=5)
VMH (n=1,225; S=4)
TU (n=1,140; S=4)
LPO (n=1,139; S=6)
MPO (n=1,102; S=6)
PVH (n=1,057; S=5)
MPN (n=1,016; S=5)
DMH (n=945; S=4)

0.00 0.25 0.50 0.75 1.00
Detection probability

B Fosl2

HY-unassigned (n=6,393; S=18)
LHA (n=4,502; S=11)
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0.00 0.25 0.50 0.75 1.00
Detection probability

C Nr4a3

HY-unassigned (n=6,393; S=18)
LHA (n=4,502; S=11)
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0.00 0.25 0.50 0.75 1.00
Detection probability

D New section, actual n

Tll1
Fosl2
Nr4a3
Egr2
Cpne9
Egr3
Hgf
Bcl11b
Car12
Crym

Nominal95%

0.0 0.5 1.0
Held-out count PI coverage

E Held-out fraction error

Tll1
Fosl2
Nr4a3
Egr2
Cpne9
Egr3
Hgf
Bcl11b
Car12
Crym

0.0 0.1 0.2
Fraction MAE
(section × anatomy)

F Grid sensitivity
full-data μ means

Tll1
Fosl2
Nr4a3
Egr2
Cpne9
Egr3
Hgf
Bcl11b
Car12
Crym

0.000 0.004 0.008
Maximum |Δ posterior μ mean|

Anatomy Beta-Binomial
Regional training-only baseline
64×48 vs 32×24
64×48 vs 96×64

One unexposed donor; physical sections model heterogeneity, not independent animals. All measured zeros retained.
Fixed μ~Beta(½,½), κ~LogNormal(log20,1.5);64×48 prior-quantile grid;8,192 seeded mixture draws. OOF prior-only groups=790/1860. No PM2.5 effect or new spatial resolution.